Reg. No. [ ][ ][ ][ ][ ][ ][ ][ ][ ][ ] [14EC42]

FOURTH SEMESTER B.TECH UNIVERSITY EXAMINATION, JUNE – 2016

DIGITAL SYSTEM DESIGN USING HDL

Time: 3 Hours **Maximum Marks:70**

Instructions:

1. **Answer any FIVE full questions choosing at least TWO questions from each Part.**
2. **Any missing data may be suitably assumed**

PART-A

1. a. Explain the following VHDL data types. 08
   - i. Physical
   - ii. User defined
   - iii. Severity
   - iv. Array

   b. Write switch-level description in VHDL for the inverter circuit with components NMOS and PMOS. 06

2. a. Bring out the differences between execution of concurrent and sequential signal assignment statements in VHDL taking suitable architectural examples. 06

   b. Draw the block diagram of a 3-bit carry-look-ahead adder. List the Boolean functions and write the data flow description for the same in Verilog. 08

3. a. Explain with suitable examples 08
   - i. FOR loop and WHILE loop in VHDL and Verilog.
   - ii. Verilog Repeat and Forever.

   b. Write the VHDL code to find the factorial of Positive integers. 06

4. a. Explain with suitable example, how binding is achieved in VHDL between 06
   - i. Entity and Architecture
   - ii. Entity and Component
   - iii. Library and Module

   b. Write mixed-language description of a Master-Slave D-Flip Flop, invoking a VHDL entity from a Verilog Module. 08

PART-B

5. a. Mention the highlights of switch level description. **04**

b. With truth table and switch-level logic diagram, write code for a Two- input NAND gate in VHDL and Verilog. **10**

6. a. With syntax, explain package declaration and package body. **04**

b. Write a VHDL code to convert unsigned binary to integer using procedure. **08**

c. List the file processing tasks in Verilog. **02**

7. a. With neat flow chart, explain the steps involved in synthesis process. **06**

b. Write a function to realize the statement $Y = 2 * a + 5$, where a is a 3-bit input, (a<=4), Y is a 4-bit output. Incorporate the same in Verilog module. Show the synthesis result of the above module. **08**

8. a. Write VHDL code for the finite sequential state machine shown in figure 8(a). **06**

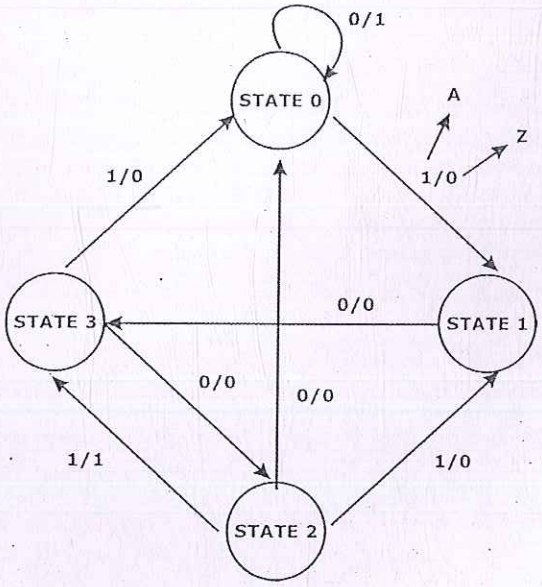

Figure 8(a)

b. Explain in brief the working of serial adder with accumulator (4-bit numbers) with block diagram. Design control circuit for the adder, arrive at the state graph and table for the same. **08**